

Problem A. K Co-prime Permutation

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Kotori is very good at math (really?) and she loves playing with permutations and primes.

One day, she thinks of a special kind of permutation named *k co-prime permutation*. A permutation $p_1, p_2, \dots, p_n$ of n is called a *k co-prime permutation* of n if there exists exactly k integers i such that $1 \leq i \leq n$ and $\gcd(p_i, i) = 1$, where $\gcd(x, y)$ indicates the greatest common divisor of x and y .

Given n and k , please help Kotori construct a *k co-prime permutation* of n or just report that there is no such permutation.

Recall that a permutation of n is a sequence of length n containing all integers from 1 to n .

Input

There is only one test case in each test file.

The first and only line contains two integers n and k ($1 \leq n \leq 10^6$, $0 \leq k \leq n$).

Output

Output one line containing n integers $p_1, p_2, \dots, p_n$ separated by one space, indicating the permutation satisfying the given constraints. If no such permutation exists output “-1” (without quotes) instead. If there are multiple valid answers you can print any of them.

Please, DO NOT output extra spaces at the end of each line, otherwise your answer may be considered incorrect!

Examples

standard input	standard output
5 3	1 4 5 2 3
1 0	-1

Problem B. Social Media

Input file: **standard input**
Output file: **standard output**
Time limit: 2 seconds
Memory limit: 1024 megabytes

On a social media platform, users can leave comments under others' posts to express their thoughts. However, these comments are not visible to everyone. Specifically, for user C to see user A 's comments under user B 's post, he/she has to be friends with both A and B at the same time. If a user leaves a comment under his/her own post, all his/her friends can see this comment.

As an active user on this platform, you would like to see as many comments as possible. There are k users (not counting you) on the platform, numbered from 1 to k . There are also m comments on the platform, but you might not be able to see them all because you only have n friends. As you need to participate in the 2024 ICPC Asia Nanjing Regional Contest, you don't have time to make too many new friends. What's the maximum number of comments you can see if you make at most two new friends on the platform?

Input

There are multiple test cases. The first line of the input contains an integer T indicating the number of test cases. For each test case:

The first line contains three integers n , m , and k ($1 \leq n \leq k \leq 2 \times 10^5$, $1 \leq m \leq 2 \times 10^5$) indicating the number of your friends, the number of comments, and the number of users (not counting you) on the platform.

The second line contains n distinct integers $f_1, f_2, \dots, f_n$ ($1 \leq f_i \leq k$) indicating your friends on the platform.

For the following m lines, the i -th line contains two integers a_i and b_i ($1 \leq a_i, b_i \leq k$) indicating a comment written by user a_i under user b_i 's post.

It's guaranteed that neither the sum of k nor the sum of m of all test cases will exceed 2×10^5 .

Output

For each test case, output one line containing one integer, indicating the maximum number of comments you can see if you make at most two new friends on the platform.

Example

standard input	standard output
5	9
4 12 7	5
5 7 3 6	1
3 6	1
2 2	1
1 4	
2 4	
1 3	
7 6	
4 1	
5 4	
1 1	
1 1	
2 1	
3 7	
2 7 6	
2 4	
1 2	
3 2	
2 5	
5 4	
2 6	
4 6	
2 6	
1 1 2	
1	
1 2	
2 1 2	
1 2	
1 2	
2 1 100	
24 11	
11 24	

Note

For the first sample test case, you can make friends with user 1 and 4.

For the second sample test case, you can make friends with user 5 and 6.

For the third sample test case, you can make friends with user 2.

For the fourth and fifth sample test cases, you don't need to make new friends because you can already see all comments.

Problem C. Evil Coordinate

Input file: **standard input**
 Output file: **standard output**
 Time limit: 1 second
 Memory limit: 256 megabytes

A robot is standing on an infinite 2-dimensional plane. Programmed with a string $s_1s_2\cdots s_n$ of length n , where $s_i \in \{\text{'U'}, 'D', 'L', 'R'}\}$, the robot will start moving from $(0,0)$ and will follow the instructions represented by the characters in the string.

More formally, let (x,y) be the current coordinate of the robot. Starting from $(0,0)$, the robot repeats the following procedure n times. During the i -th time:

- If $s_i = \text{'U'}$ the robot moves from (x,y) to $(x,y+1)$;
- If $s_i = \text{'D'}$ the robot moves from (x,y) to $(x,y-1)$;
- If $s_i = \text{'L'}$ the robot moves from (x,y) to $(x-1,y)$;
- If $s_i = \text{'R'}$ the robot moves from (x,y) to $(x+1,y)$.

However, there is a mine buried under the coordinate (m_x, m_y) . If the robot steps onto (m_x, m_y) during its movement, it will be blown up into pieces. Poor robot!

Your task is to rearrange the characters in the string in any order, so that the robot will not step onto (m_x, m_y) .

Input

There are multiple test cases. The first line of the input contains an integer T indicating the number of test cases. For each test case:

The first line contains two integers m_x and m_y ($-10^9 \leq m_x, m_y \leq 10^9$) indicating the coordinate of the mine.

The second line contains a string $s_1s_2\cdots s_n$ of length n ($1 \leq n \leq 10^5$, $s_i \in \{\text{'U'}, 'D', 'L', 'R'}\}$) indicating the string programmed into the robot.

It's guaranteed that the sum of n of all test cases will not exceed 10^6 .

Output

For each test case output one line. If a valid answer exists print the rearranged string, otherwise print "Impossible" (without quotes) instead. If there are multiple valid answers you can print any of them.

Example

standard input	standard output
5	LDLRUUR
1 1	UUU
RURULLD	Impossible
0 5	Impossible
UUU	Impossible
0 3	
UUU	
0 2	
UUU	
0 0	
UUU	

Problem D. Chat Program

Input file: **standard input**
Output file: **standard output**
Time limit: **3 seconds**
Memory limit: **512 megabytes**

You're the researcher of the *International Chat Program Company* (ICPC). Today, you discover the following chat history when reviewing some research data.

SUA (2022/12/04 23:01:25)

I'm out of ideas for competitive programming problems! Please give me a problem about sequences.

BOT (2022/12/04 23:01:27)

Sure. Here is a competitive programming problem about sequences.

Given an integer sequence $a_1, a_2, \dots, a_n$ of length n and four other integers k, m, c and d , your goal is to maximize the k -th largest element in the sequence.

To achieve the goal, you can perform the following operation at most once: select a continuous sub-array of length m and add an arithmetic sequence with length m , initial term c and common difference d to the sub-array.

More formally, you can select an integer p satisfying $1 \leq p \leq n - m + 1$ and add $(c + di)$ to a_{p+i} for all $0 \leq i < m$.

Calculate the largest possible value of the k -th largest element in the sequence after at most one operation.

The k -th largest element in the sequence is the k -th element in the sorted sequence after sorting all elements from the largest to the smallest. For example, the 3rd largest element in sequence $\{5, 7, 1, 9\}$ is 5, while the 3rd largest element in sequence $\{9, 7, 5, 9\}$ is 7.

SUA (2022/12/05 00:15:17)

This problem seems difficult! Please teach me the solution.

BOT (2022/12/05 00:15:30)

Sure. Firstly, we can...

[DATA EXPUNGED]

Unfortunately, parts of the chat history are lost due to a disk failure. You're amazed at how a chat program can create a competitive programming problem. To verify whether the chat program can create valid problems, you decide to try on this problem.

Input

There is only one test case in each test file.

The first line contains five integers n, k, m, c and d ($1 \leq k, m \leq n \leq 2 \times 10^5$, $0 \leq c, d \leq 10^9$) indicating the length of the sequence, your goal, the length, initial term and common difference of the arithmetic sequence.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$) indicating the sequence.

Output

Output one line containing one integer indicating the largest possible value of the k -th largest element in the sequence after at most one operation.

Examples

standard input	standard output
6 4 3 1 2 1 1 4 5 1 4	4
7 3 2 4 0 1 9 1 9 8 1 0	9
8 3 5 0 0 2 0 2 2 1 2 1 8	2

Note

For the first sample test case, we can choose $p = 3$ so the sequence becomes $\{1, 1, 5, 8, 6, 4\}$. The 4-th largest element in the sequence is 4.

For the second sample test case, we can choose $p = 5$ so the sequence becomes $\{1, 9, 1, 9, 12, 5, 0\}$. The 3-rd largest element in the sequence is 9.

For the third sample test case, it is easy to see that the operation does not change the sequence, so we choose not to perform the operation. The 3-rd largest element in the sequence is 2.

Could you encourage the competitors of a competitive programming contest?



Sure, here are a few words of encouragement for competitors in a competitive programming contest:



- Keep up the hard work! You've put in a lot of effort to get to this point, and it's important to keep pushing yourself.
- Don't be afraid to ask for help. If you're stuck on a problem, don't hesitate to reach out to others for guidance. → *only your teammates!*
- Stay positive. It's natural to feel frustrated when you're facing a difficult problem, but try to stay positive and keep moving forward.
- Take breaks when you need to. It's important to stay focused and avoid burnout. If you're feeling overwhelmed, take a break and come back to the problem later with a fresh mind.
- Remember that everyone has their own strengths and weaknesses. Don't compare yourself to others – focus on your own progress and improvement.

Overall, the most important thing is to have fun and enjoy the challenge. Good luck!

OpenAI ChatGPT is encouraging the contestants